# DA05 Lists, dictionaries and loops

Week 2, Python you need | Session 5 of 24 | Python for Data Analysis

**Today's question.** How many units did the first six sales sell in total?

**Objective.** By the end: you can store several values in a list, look one up by position, and use a loop to add them up.

> **WHERE THIS IS USED**
> A shop has many sales, a league has many matches and a tailor has many orders. A loop handles every one of them without you typing each by hand. What would you loop over in your favourite sport?

Before this session: DA04 Conditions and decisions. If you cannot explain the last example, use its error card with the instructor first.

## Words for today

| Word | What it means | Think of it like this |
|---|---|---|
| list | Several values in order inside square brackets. | A shopping list. |
| index | The position of a value in a list. Counting starts at 0. | The numbered seats in a bus, with the driver as seat 0. |
| dictionary | Named pairs inside braces. A key finds its value. | A price board: the name of the item gives its price. |
| key | The name used to look up a value in a dictionary. | The name on a price board. |
| loop | Repeat the indented lines once for each value. | Counting stock shelf by shelf. |
| running total | A number that grows as each value is added. | The cash tin during the day. |

## Block A. Lists and positions

The units of the first six sales in the shop table are 1, 2, 4, 7, 6 and 5. A list keeps them in order inside square brackets. Square brackets after a list name pick out one position. Position 0 is the first.

**Predict.** Predict what each print shows. Write your guess on paper first.

*Your guess, written before you run it*

```
units = [1, 2, 4, 7, 6, 5]
print(units[0])
print(units[3])
print(len(units))
print(max(units))
```

**Run it, then compare.** The cell prints 1, 7, 6 and 7. Position 0 holds the first value and position 3 holds the fourth. len counts the values and max finds the largest.

**Investigate.**

1. Why does units[3] show 7 and not 4?
2. Which position holds the last value?

**Modify.** Print the value at position 4.

## Block B. Dictionaries and a loop with a running total

A dictionary stores named pairs. A for loop repeats its indented lines for each value in a list. The name after for stands for the current value.

**Predict.** Predict the final total and the price of Pen. Write your guess on paper first.

*Your guess, written before you run it*

```
units = [1, 2, 4, 7, 6, 5]
prices = {"Book": 3000, "Pen": 1000, "Juice": 2000}
print(prices["Pen"])
total_units = 0
for u in units:
    total_units = total_units + u
print(total_units)
```

**Run it, then compare.** The cell prints 1000 and then 25. The loop starts the running total at 0 and adds each of the six values in turn: 1, 3, 7, 14, 20, 25.

**Investigate.**

1. What does the line total_units = total_units + u do?
2. What would happen if total_units = 0 sat inside the loop?

**Modify.** Change the loop to print each value plus 1 and then print the final total.

## Make

Make a list [2, 4, 6]. Print each value plus 1 with a loop. Make a dictionary with the keys name and stock and print the stock. Explain why position 0 is the first.

Save your work as DA05. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

## Error card

| You see | It means | Fix |
|---|---|---|
| IndexError: list index out of range | You asked for a position beyond the end of the list. | A list of six values has positions 0 to 5. Check len(). |
| KeyError: 'Price' | The key is not in the dictionary, often because of a capital letter. | Check the exact spelling and capitals of the key. |

## Exit ticket

A list keeps values in order starting at position 0, and a loop repeats the same steps for each value.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.